

Assignment 5: Orthogonality and Determinant(Lec16 to Lec20)

Assigned: Apr 6, 2026 Monday

Due: Apr 15, 2026 Wednesday at 10:00pm

Rules (please read!!)

- **English:** Answer the questions in English. Otherwise, you'll lose half of the points.
- **Electronic submission:** Turn in solutions electronically via Blackboard. Be sure to submit your homework as a single file.
- **Collaboration policy:** Collaboration is allowed for all problems, but please list all the people with whom you discussed. Crediting help from other classmates will not take away any credit from you. However, only insightful discussions are allowed. Directly sharing the solutions is prohibited. The details of the collaboration policy for this course are available in the Resources tab on Piazza.
- **Questions on homework:** Start early and come to TA office hours with your questions on the assignments!

TA Email: Xingyu WANG 225040497@link.cuhk.edu.cn

Total: 100 points, 6+1 problems.

1 Least Squares Problem (20 points)

The following table shows the temperature data measured at four time points:

Time t (hours)	0	1	2	3
Temperature y ($^{\circ}\text{C}$)	2	5	9	14

Suppose the relationship between temperature and time is $y = at + b$. Form a least squares problem, find the best estimates for parameters a and b , and predict the temperature at $t = 4$.

2 Orthogonality (10 points)

Suppose S is a subspace of $\mathbb{R}^n$. Suppose $\mathbf{b} \in \mathbb{R}^n$ and $\mathbf{p} \in S$ satisfy $(\mathbf{b} - \mathbf{p}) \perp S$. Prove that for any $\mathbf{z} \in S$, we have

$$\|\mathbf{b} - \mathbf{z}\|^2 - \|\mathbf{b} - \mathbf{p}\|^2 = \|\mathbf{z} - \mathbf{p}\|^2.$$

3 Orthonormal Basis (20 points)

Given the set of vectors:

$$\mathbf{u}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \quad \mathbf{u}_2 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{u}_3 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

Let $W = \text{span}\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\} \subseteq \mathbb{R}^4$.

(a) Use the Gram-Schmidt Process to construct an orthonormal basis $\{\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3\}$ of W .

(b) Find the projection of vector $\mathbf{v} = \begin{bmatrix} 2 \\ 0 \\ 0 \\ 0 \end{bmatrix}$ onto W .

4 Determinant (15 points)

Let $\mathbf{A}_n$ be the following $n \times n$ tridiagonal matrix:

$$\mathbf{A}_n = \begin{bmatrix} 2 & 1 & 0 & \cdots & 0 \\ 1 & 2 & 1 & \ddots & \vdots \\ 0 & 1 & 2 & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & 1 \\ 0 & \cdots & 0 & 1 & 2 \end{bmatrix}$$

Let $D_n = \det(\mathbf{A}_n)$.

(a) Prove that $D_n = 2D_{n-1} - D_{n-2}$ for $n \geq 3$.

(b) Prove: $D_n = n + 1$. Hint: Analyze D_1 and D_2 first, and then apply the formula.

5 Determinant (15 points)

Prove:

(a) Let $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{n \times n}$ and suppose $\mathbf{A}$ is invertible. Prove that $\det(\mathbf{A}^{-1}\mathbf{B}\mathbf{A}) = \det(\mathbf{B})$.

(b) Let $\mathbf{A}$ be an $n \times n$ orthogonal matrix with $\det(\mathbf{A}) = 1$, and let $\mathbf{B}$ be any $n \times n$ matrix. Prove that $\det(\mathbf{A}^T\mathbf{B}\mathbf{A}) = \det(\mathbf{B})$.

(c) Let $\mathbf{A}$ be a skew-symmetric matrix in $\mathbb{R}^{2025 \times 2025}$, i.e., $\mathbf{A}^T = -\mathbf{A}$. Prove that $\det(\mathbf{A}) = 0$.

6 Determinant via Laplace Expansion (10 points)

Consider the matrix

$$\mathbf{A} = \begin{bmatrix} 0 & 2 & 3 & 0 \\ 0 & 4 & 5 & 0 \\ 0 & 1 & 0 & 3 \\ 2 & 0 & 1 & 3 \end{bmatrix}.$$

- (a) Compute $\det(\mathbf{A})$ using Laplace expansion along the first column.
- (b) Verify your result by computing $\det(\mathbf{A})$ using Laplace expansion along the fourth column.

7 Bonus Problem

Set

$$\theta = \frac{2\pi}{n}, \quad A = (a_{s,t})_{1 \leq s,t \leq n}, \quad a_{s,t} = \sin((s+t)\theta).$$

Compute $\det(I_n + A)$.

To guide the proof, answer the following questions.

(a) Define vectors $u, v \in \mathbb{R}^n$ by

$$u_s = \sin(s\theta), \quad v_s = \cos(s\theta), \quad s = 1, \dots, n.$$

Show that

$$A = uv^T + vu^T.$$

(b) Treat separately the cases $n = 1$ and $n = 2$. Compute $\det(I_n + A)$.

(c) Now assume $n \geq 3$. Let

$$U = [u, v], \quad V = [v, u],$$

so that $A = UV^T$.

You may use, without proof, the following block determinant formulas:

$$\det \begin{pmatrix} A & B \\ C & D \end{pmatrix} = \det(D) \det(A - BD^{-1}C) \quad \text{if } D \text{ is invertible,}$$

$$\det \begin{pmatrix} A & B \\ C & D \end{pmatrix} = \det(A) \det(D - CA^{-1}B) \quad \text{if } A \text{ is invertible.}$$

Apply them to show that

$$\det(I_n + A) = \det(I_2 + V^T U) = (1 + u^T v)^2 - \|u\|^2 \|v\|^2.$$

(d) Compute $u^T v$, $\|u\|^2$, and $\|v\|^2$, and calculate $\det(I_n + A)$ for all cases of n .